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Application Scorecard Modelling

Techniques and Performance

by

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Under the guidance of

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Submitted for degree requirement of MSc Finance

VU University Amsterdam

October 2016

APPLICATION SCORECARD MODELLING:
TECHNIQUES AND PERFORMANCE

SUMMARY

This research aims to review and assess methods for improving the predictive power of credit scoring models for Auto-Leasing business. We examine the process-flow involved in the scorecard modeling and highlight the probable cause of weak model performance. Such examination would include theoretical and empirical assessment of data quality, variable transformation and selection, statistical tests for model performances and so on. Moreover, we aim develop an application scorecard based on automated binning algorithms and assess its validity and practicality for implementation.

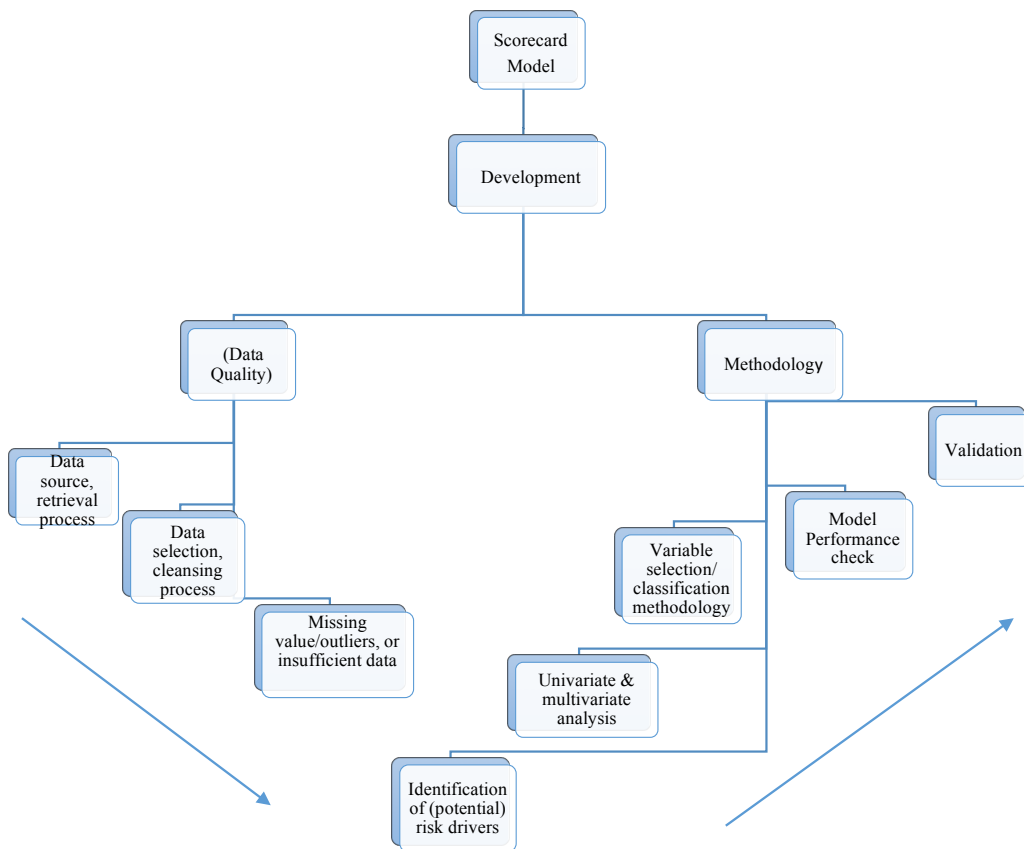


Figure 1: Diagram of Modeling Process

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Chapter 1

Introduction

In this research, we focus on application scorecard development process for assessing credit risk of retail businesses and small medium enterprises (SMEs). According to Allen et al. (2003), academic researchers had been mainly focused bank risk models, and meanwhile the area of retail credit risk remained relatively underdeveloped. This was mainly due to the lack of available data to the researchers. Moreover the retail loans and credit sector had shown slower progress in terms of development of credit risk models. Gradually, the retail sector has become of interest to regulators and researchers due to it's growing importance with increasing volumes of outstanding credit.

Construction and maintenance of a scorecard model require quite a number of stages, including data preparation, model development, training & validation, calibration, implementation, overrides & controls, and monitoring (Anderson, 2007). Given the tediousness of completing all the stages, we focus on only particular parts of the model development process.

The primary influences in the modelling process are:

1. The amount of available data
2. Business usage and modelling purpose
3. Available techniques and platforms

(1) Our data was provided by LeasePlan¹ entity in Italy (LPIT) and we developed several models whose performances are bench-marked by current LPIT application scorecard. (2) The scorecards are developed for the purpose of automated approve/reject decisions for the credit portfolio of retail businesses and small medium enterprises. (3) We used R programming language for our scorecard development process due to its wide variety of available graphical and statistical programming features.

¹LeasePlan is an international company of Dutch origins specialized in fleet management.

In **Chapter 1**, we start our research with basic background check on credit scoring models. Then, we move to the topic of scorecard development process in **Chapter 2**. In section 2.1 we mainly focus on data quality issues and handling of missing values in data set. Decent quality of data is of main importance since the scorecard performance is heavily dependent on the underlying data. Number of techniques have been suggested by researchers to deal with the missing values in data, including depletion and imputation techniques. When the missing rate in the data set is large, depletion technique has a danger of leading to information loss. Also, a selection of proper imputation technique must be considered with utmost caution since it can lead to serious bias. Alternatively, we treated the missing values in a particular predictor variable by placing its missing values into separate bucket/class. This separate class approach is mainly used in credit scoring models through the variable transformation and binning process (Fleming, 2013).

For the rest of the Chapter 2, we review popular modelling techniques for scorecards and their suitability for business implementation. It's important for an application scorecard to have interpretable outcome, so that the decision logic can be understood, where the companies must advise decision reasons to customers (Anderson, 2007). Decision trees based on simple recursive partitioning algorithms provide easy to interpret outcomes, however their performance appear to be modest due to their unstable nature. Boosted tree and Neural Network models perform relatively well, however at the expense of interpretability. Therefore, we proceed our rest of research building on logistic regression models, given their ease of interpretability and acclamatory robust performance.

In most cases the attributes of predictors are segmented into grouping intervals, with the aim of creating bins for the model that maximize the correlation with these attributes (Oliveira et al., 2008). Variable transformation through binning is an essential part of logistic regression model development process as the characteristics of bins, measured by weight-of-evidence (WoE), impacts the predictive power of credit scoring model. Proper binning of a predictor variable is subject to certain constraints, including min/max number of bins, min/max percentages of observations per bin, and monotonicity of aggregate bin WoE values etc. Such constraints and assumptions, that depend on availability and nature of the data, can make the binning process tedious, especially when enforced manually and when the number of predictors are large. Automated binning processes can drastically reduce the time and efforts spend on scorecard development process and reduce the risk of eliminating significant variables when using variable reduction techniques. Therefore, in **Chapter 3** we review possible algorithms for automated

binning process, and their underlying techniques and theories.

We built several logistic regression models based on these algorithms and assessed their performances in **Chapter 4**. Our proposed automated binning model, **MAPA+Clust**, which employs Monotonic Adjacent Pooling Algorithm in combination with cluster analysis, effectively bins the predictor variables and outperforms the benchmark model. Finally, **Chapter 5** concludes with a summary of the key findings in this research work.

1.1 *Background: Credit Scoring Models*

Perhaps the simplest way to explain credit-scoring is to separate the word components into "credit" and "scoring", where the word *credit* represents borrowers willingness and ability to pay back, as well as the potential financial impact due to observed or perceived changes in borrower's creditworthiness. Meanwhile, the word "scoring" implies ranking and ordering of cases or borrowers according to their perceived or observed quality which may distinguish into the separate groups of good and bad borrowers. Hence, credit scoring is the use of statistical models and techniques to translate relevant data of into numerical scores that guide credit decisions. Generally, higher the estimated score, less the riskiness of the client, and thus more reliable the client is perceived.

In general, credit portfolio of banks or financial institutions consist of business (small, medium, large, retail, ect.) portfolios and individual(private clients) portfolios. Perhaps, creditworthiness of larger companies may assessed based on the ratings of external agencies, such as Standard & Poor's, Moody's or of credit bureaus. Those external ratings, however, are mostly not available for individual clients, retails or small medium enterprises, and hence the applications are required to go through internal screening.

Scorecard models can be described as *type I* model and *type II* model, such that type I models are used as an input for PD and LGD models, whereas type II models are application scorecard for approval or reject decision making. Application scorecards are well-suited in high-volume and low-value environments, and they replace the traditional approach to credit assessment based on expert judgements and reduces the time of application to be processed through the credit committee. As we boil down all the concepts into a single sentence, the main principle of credit scoring system is to be an optimal classifier which can assign a score to each borrowers in order to differentiate the bad clients from the good clients.

1.2 Application Scorecards for Auto-leasing Business

Application credit-scoring models for leasing slightly differ from the models developed for bank lending. According to Caire (2005), bank lending scorecards mainly concern in quantifying an applicant's likelihood of default on a credit obligation. Models for leasing assets also aims to assess the client's default risk, but with additional considerations embedded, including the asset worth and vendor relationships, which affect the overall risk of the lease contract. Here below, we mention the three most important differing factors in accordance with Caire (2005):

- Banks take pledge against the financed assets, and perhaps some additional assets for collateral purpose. Meanwhile, lessors (leasing companies) are entitled to the financed asset as they legally owns them.
- Leasing companies work closely with particular selection of vendors that usually provide buy-back guarantees. Such guarantees can mitigate the risks of bad deals.
- Most leasing companies offers financing packages for standard equipments that generally have liquid secondary markets, such as automobiles and vehicles.

These intrinsic differences between loan and leasing risk imply that the leasing models should focus on not only on the client's inherent default risk, but also on the expected loss or profit from asset sale in the event of default. Consequently, such kind of additional consideration may require longer development process for the model. Thus, more we automate the modelling process, faster it is to process additional factors and easier it is to validate the model; i.e. automated binning process.

1.3 Literature Review & Research Contribution

We believe that our research findings contribute to the field of credit scoring in two ways:

1. *We focus on automated binning process.*

One may find extensive collection of researches on variety of topics related to credit scoring models. However, we believe that automated binning process is one of the least explored subject among the previously written literatures. The most elaborate explanations on techniques and algorithms of binning process can be found in textbooks like *The Credit Scoring Toolkit* by Anderson (2007), *Business Modelling and Data Mining* by Pyle (2003), and *Credit Scoring and Its Applications* by Thomas et al. (2002). Another useful sources are user guidelines and documentations on automated binning tools implemented commercial statistical softwares like SAS and Matlab, as well as binning packages developed on

open sources like R and Python (see Refaat, 2007; Jopia, 2016; Oliveira et al., 2008) . One should also take a look at Zeng (2014) for theoretical derivations of necessary conditions for a good binning algorithm. In this work, we review and assess performances of popular binning algorithms and tools, in comparison with manual binning based on expert knowledge. We believe such assessment can be an additional contribution.

2. *We propose a new binning method for nominal variables: WoE based Cluster Binning Algorithm*

Automated binning algorithms based on clustering of homogeneous attributes of predictor variable very common in the field of pharmacometrics². Decent examples can be found from Sonehag et al. (2012) and Nugent and Stuetzle (2010), where they employ cluster analysis for selection of the number of bins for simulated data on drug uptake, effects and elimination. However, these pharmacometrics models simply measure the lasting effect of drug in human body over time, rather than accounting for any related risk or side effects. Moreover, cluster analysis is an unsupervised learning which does not account for any relationship between the predictor and target variables. On the other hand, it's essential for the credit risk models that characteristics of bins to be representative of default risk, i.e. higher income class clients should be less risky than lower income class clients (O.Duda et al., 2000). Due to this reason, the binning approach based on cluster analysis has not been explored yet, or scarce literatures can be found if there's any, in the framework of scorecard modelling. To account for the relationship between target variable and a predictor variable, we cluster the factor observation based on their Weight-of-Evidence (WoE). Similar binning suggestions based on WoE, such as collapsing adjacent attributes with similar WoE scores is demonstrated in (Bhalla, 2015).

²Mathematical modelling of biology, pharmacology, and physiology used to quantify interaction between drug intake and patients.

Chapter 2

Scorecard Development

2.1 The Data

2.1.1 Data Quality

Successful performance of a scorecard depends mostly on the data used during the development stage. Therefore, the first issue that needs to be addressed is the quality of available data. Data quality issues mainly concern with:

- *accuracy*
- *consistency*
- *completeness* of the data set (Baesens et al., 2003; Lindsay et al., 2010)

Perhaps, common example accuracy issue in the data relates to the user input errors or errors in software. Data completeness refers to which values are missing in the data (Parker et al., 2006). Data consistency relates to situations in which multiple data sources are used and due to lack of standardisation, where data items may conflict with each other.

Moreover, sufficiency in the quantity of data is required to ensure construction of good quality and robust scorecard models. Usually in application scorecard modelling, it's recommended to use 1,500 instances of each class, where reject inference is performed and an additional 1,500 rejected applicants are required.

Validity of this recommendations is based on the understanding that the composition of credit scoring data set is homogeneous across lenders and regions. For example, consider the different sources of data construction (Crone and Finlay, 2012):

- Internal : This type of data offers insight into the behavioural accounts of the client.
- External: This type of data is obtained from application forms, financial statements etc.

- Bureau : This relates to data recorded by credit bureau and court instances. Credit bureau are institutions that collect data on the performance of loans granted by a different lenders.

Finally, the dataset should contain enough instances, including the default instances, to restrict the occurrence of correlated variables which result on over-fitting of the credit-scoring model.

2.1.2 Missing Values

Dealing with missing values in our data set was one of the serious issues we confronted with in this work, as the missing data can seriously bias credit scores. We considered number of possible techniques, including the multiple imputation techniques. But, before the selection of appropriate approach to missing data, one keep in mind that the suitability of the approach is dependant on underlying missing data mechanism of the respective data (Galler and Kehrel, 2009).

Mechanism of Missingness:

- Missing Completely at Random (MCAR)
- Missing at Random (MAR)
- Missing Not at Random (MNAR)

According to Rubin's classification (1976), an indicator variable R shows which values are observed and which are missing, and the probability distribution of R is the mechanism of missingness. Then, the complete data are divided into an observed part and a missing part:

$$Y_{com} = Y_{obs} \cup Y_{miss}; \quad Y_{obs} \cap Y_{mis} = \emptyset \quad (2.1)$$

In the case of *MCAR*, the mechanism of missingness neither depends on the observed nor on the missing data, such that there is no link between the probability of a value to missing and the data:

$$P(R|Y_{com}) = P(R) \quad (2.2)$$

In the case of *MAR*, the mechanism of missingness is independent of the missing data (Y_{miss}) and only depends on the observed values (Y_{obs}).

$$P(R|Y_{com}) = P(R|Y_{obs}) \quad (2.3)$$

However, in the case of *MNAR*, the probability distribution of R depends also on the missing data, meaning that the mechanism is additionally dependent on the values of the variable which contains the missing data:

$$P(R|Y_{com}) = P(R|Y_{obs}, Y_{mis}) \quad (2.4)$$

In reality, data corresponding to MCAR are rarely occurs. Many methods for dealing with missing data requires for MAR. Once the mechanism of missing value is detected, one may consider proper method of treatment. A very common case is *list-wise depletion*, where only fully available units¹ are subject to further analysis while the rest is discarded. An disadvantage of this method is loss of available data and hence a reduction of statistical power which could be particularly problematic if data records are scarce to begin with.

On the other hand, units containing missing data do not necessarily have to be discarded. The missing values can be replaced by plausible values which helps to save the available data. This approach is called *imputation* methods. Because the filled in data points are not the real data, the quality of these methods heavily depend on the quality of the imputations. An obvious advantage if such approach is no values are dropped, so that the danger of loss of statistical power is reduced. On the other hand, this approach can entail the danger of altering the data's properties and distribution considerably (Galler and Kehrel, 2009). For example, let's consider the *mean imputation* approach, which uses the mean of the observed data of a particular variable as a substitute for the missing positions in the variable. The idea of this concept is that the mean is the most probably observed value in the normal distribution and thus resembles the best estimate of a missing value under such distribution. Since the distribution mean is inserted to the data, its average can be sustained. However, other properties of the distribution, such as variances and quantiles can be seriously distorted. The main problem resulting in the context of variances is that they are underestimated proportionally to the percentage of occurring missing data. Also, covariances and correlations between variables are altered and underestimated because the imputed values exhibit variance of 0. One may also consider *median imputation*, which is more suitable approach when dealing with data which are not symmetrically distributed. This approach is also less sensitive to individual outliers.

Both mean and median imputation approaches are examples of *single imputation* methods. Alternatively, *multiple imputation* generate m complete data sets rather than imputing a single value for all the missing data. Each of the m data set is analyzed with statistical procedure, such as logistic regression, as if the data were complete. To obtain a point estimate of a parameter, m parameters are estimated and combined. Moreover, in order to generate multiple imputations, values for the missing data are drawn from an appropriate distribution. In comparison to the list-wise depletion method, the existing data are used more efficiently, and in comparison to single imputation methods, uncertainty about missing data when estimating certain quantiles are reflected by using m draws. It's standard implementation may provide unbiased inference

¹units which exhibit observed values for every single variable

when the underlying mechanism is missing at random (MAR). However, the process of multiple imputation is complex and in the presence of missing data generated by missing not at random (MNAR) mechanism it can lead to biased inference (Galimard et al., 2016).

Alternatively, Twala et al. (2008) proposed a method related to creating a separate class for missing values, and found that its performance was competitive with that of multiple imputation methods, including likelihood-based multiple imputation. Ding and Simonoff (2010) also found in their study of several imputation techniques for classification tree models, found that separate class method tend to perform better than other alternatives, when the missing rate is neither too high nor too low, trading off between information loss due to missing values and information gain from the informative missing data generating process.

Although, couple of variable in our data set contains high rate of missing values, we assume that the most suitable method is to treat the missing values as separate class through proper coarse classification, given that this work mainly focus binning of the predictor variables. Later, in Chapter 4, we'll introduce our automated binning models. Each of these models creates separate bin class for the missing values in each predictor variables.

2.1.3 Reject Inference ²

When constructing scorecard, the outcome value is only available for clients who were actually approved/granted for credit or leases. For declined clients, as they deemed to represent a default risk, only has their characteristic values not their characteristic values but not their outcome data. This creates a selection bias, where the data set is not representative of the thought-the-door application population.

Reject inference attempts to mitigate this bias by estimating how rejected applicant would have performed if they had been accepted (Hand and Henley, 1997). In general, practitioners attempt to improve discriminatory power of the scorecard, and improve the model performance on actual application population to which it will be applied, "through-the-door" applicants. The simplest approach to the problem is to obtain the customer features and outcome for entire applicants by granting credit to every applicants during some time period. However, this is financially undesirable given the losses that are likely to occur. Alternatively, one may use commonly used reject inference approaches like random supplementation, augmentation, extrapolation, and cohort performance etc.

On the other hand, the need for reject inference varies, depending upon the current process,

²In this section, we provide brief discussion on reject inference as a part of model development and performance optimization process. However, we'll not further explore the topic as our development sample contains rejected observations, which approximately accounts for 10% of the total observations.

data, and reject rate. According to Anderson (2007), most of the literature focuses on presenting reject inference techniques, and very few attempts have been made to quantify the benefits. In his work on *Scoring Toolkit*, he stated that from the little bit of research available, there seems to be general agreement, that the potential gains from reject inference are modest. Also according to Crook and Banasik (2004),

Useful implementation of reject inference seems to depend on accurate estimation of the potential good-bad ratio for the population of all applicants. Simple application of that ratio seems indicated. More elaborate tweaking of a vast set of coefficients does not seem to promise much potential benefits.

2.2 Modelling Framework

In this section, we provide an overview of popular predictive modelling techniques for credit scoring. We attempt to describe the both parametric and non-parametric modelling techniques. Each of the methods has its own strengths and weakness, which often vary according to the circumstances and depend on the data quality.

Table 2.1: Predictive Statistics Overview

Method	Main technique	P/ NP	Summary
Logistic regression	Maximum likelihood estimation	P	Determine formula to estimate binary response variable.
Decision trees	Recursive Partitioning Algorithms	NP	Uses tree structure to maximize between group differences.
Neural Networks	Multilayer perceptron	NP	Artificial Intelligence technique, whose results are difficult to interpret.

2.2.1 Logistic Regression Model

Our starting point is the most commonly used parametric method for credit scoring. Logistic regression uses Maximum likelihood estimation process, which transforms the dependent variable into a log function and estimates the regression coefficients in a way it maximizes the log-likelihood. According to Garson (2005), it is an iterative process, which may require about six attempts to reach convergence, using a proper convergence criteria. The end result of logistic

regression can be formulated as follows:

$$\text{LogitRegression} : \ln\left(\frac{p(\text{Good})}{1 - p(\text{Good})}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + e = Z \quad (2.5)$$

The left-hand side of this equation is the natural logarithm of the odds. Natural logarithm has a nice property, which makes the equation easy to convert back into probability:

$$p(\text{Good}) = 1 - \frac{1}{1 + \exp(Z)} \quad (2.6)$$

Logistic regression requires the following assumption:

- The target variable is categorical (binary in the case of credit scoring)
- It has linear relationship with the log odds function
- It has independent error terms
- The predictor variables are uncorrelated (Mays, 2004).

Logistic regression has been an optimal choice for developing credit scoring models as it's designed to handle binary outcome and it provides final probability that cannot fall outside of the range 0 to 1. Also, it provides fairly robust estimate of the true probability, given available information (Anderson, 2007).

2.2.2 *Decision Tree Models : Pruned & Ensemble Tree Models*

Recursive partitioning offers an alternative to parametric methods. Non-parametric approaches are appealing in a sense that it does not require, or few if any, assumptions about the underlying data, compared to parametric methods.

Decision tree models utilize recursive partitioning algorithms to produce terminal nodes that are homogeneous. The top of a tree is referred as the root node, each subsequent level as a daughter node, and the bottom ones are called the terminal nodes. The root node contains a sample of subjects from which the tree is grown in the learning/training sample. All nodes in the same layer constitute a partition of the root node. Moreover, every node in the tree is merely a subset of the learning sample.

Splitting & Stopping rules, and Tree Pruning For an ordinal predictor variable, the number of allowable splits is one fewer than the number of its distinctly observed values. Similarly, any nominal/categorical variable that has k levels contributes to $2^{k-1} - 1$ allowable splits, but no intrinsic ordering to the attributes. The selection of one or several preferred splits from the pool

of all allowable splits can be set by a predefined selection criterion. Such selection criterion, also referred as *splitting rule*, is required to split the population into homogeneous groups which are mutually exclusive to each other. The main consideration is to minimize the distance between the attributes in the same group, which is based on similar default rates, and maximize the distance between the groups. In general, Gini index, impurity index, or Kolmogorov-Smirnov (KS) statistic are used as a measure of the selection criterion.

The recursive partitioning process may proceed until the tree is saturated in a sense that the offspring nodes are subject to no further division. However, the saturated tree is usually too large to be useful, as a result the terminal nodes are too small to make a sensible statistical inference. Therefore, it's preferable to set a minimum size of a node as a priori. We call it the *stopping rule*. Depending on the stopping threshold, the partitioning tends to end too soon or too late. An alternative way of fixing the over-fitting problem of saturated tree is through *pruning*. This process involves in finding a sub-tree of the saturated tree that is the most "predictive" of the outcome and least vulnerable to the noise in the data.

When the final tree is established, the terminal node values could be used either as estimates of scores, or as a grouping tool. As Anderson (2007) explained,

For a binary outcome, the values is a probability, say $P(\text{Bad})$, and all branches with the probabilities beyond a predefined cut-off, say above the average $P(\text{Bad})$, would be put into the Bad group.

(Refer to **Figure 2.1** for graphical illustration)

Ensemble of Trees, Bootstrap Aggregation, and Boosting Traditional tree-based methods are simple and easy to interpret. However, they appear to be not very competitive with the widely used supervised learning approaches like regression models, in terms of prediction accuracy due to their unstable nature. Therefore, we also discuss the ensemble of trees methods. These kind of methods involves in growing multiple trees which are then combined to perform a single consensus prediction.

Bootstrap aggregation, or *bagging* in short, is one example of ensemble tree methods. It is a procedure for reducing the variance of a statistical learning. We can bootstrap by taking repeated samples from the training data set, grow a tree for each of the boosted samples, and then average all the predictions to obtain the final bagging result. This statement can be summarized by the following equation:

$$\hat{f}_{bag}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}^{*b}(x) \quad (2.7)$$

Where B is the number of bootstrapped training data sets and $\hat{f}^{*b}(x)$ is the prediction of the b th bootstrapped training set at a point x . This way, we average the predictions of large number

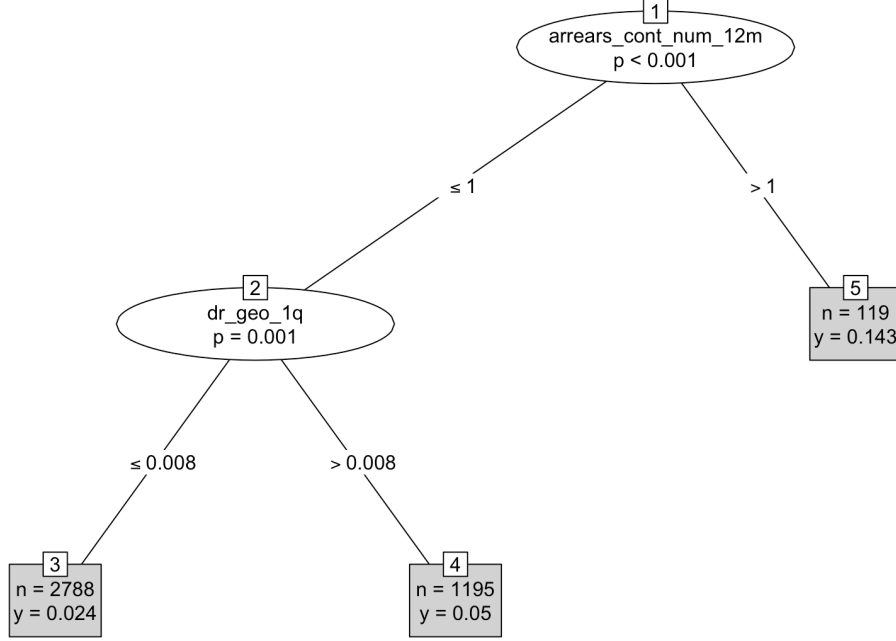


Figure 2.1: Decision Tree. Clients' record of arrears in 12 months has been selected as the root node and the optimum split was determined at the record of 1. Then, we see further split on delinquency rate of geographical area in one quarter at optimum split rate of 0.008. Notice that the p-value for the second split is slightly higher than the first split. This is due to the predefined splitting rule, where nodes split only if the null hypothesis of independence between a response and the predictors cannot be rejected at the given level significance (1-minicriterion). In the grey boxes, n represents the number of observations in the node bucket and y represents the corresponding default probability. Notice that $P(Bad)$ increases with the delinquency rate and arrears records.

of trees and reduce the variance, hence the overall prediction is the most commonly occurring class among the B predictors.

Like bagging, *boosting* is a general approach that can be applied to many statistical methods and decision trees. Boosting works in a similar way to bagging, except that it's a bit more complex and the trees are grown sequentially. This approach does not fit a single large decision tree to the data but learns slowly in a way that the subsequent trees fit to the residuals from the fitted function of the preceding tree. Each of the trees would be quite small, with just a few predefined d terminal nodes. As we fit marginally decreasing small trees to the residuals, and repeatedly updating the residuals, we slowly improve the fitted function $\hat{f}$ in areas where it does not perform well. To provide better understanding, we borrow the following algorithm on boosting (see James et al., 2013):

1. We start with $\hat{f}(x) = 0$ and $r_i = y_i$ for all i in the training set.
2. For each of B bootstrapped training set, we repeat the following process:

- (a) We fit a tree $\hat{f}^b$ with d splits to the training data (X, r) .
- (b) We update $\hat{f}$ by adding in a shrunk version of the new tree: $\hat{f}(x) \leftarrow \hat{f}(x) + \lambda \hat{f}^b(x)$, where λ is the shrinking factor.
- (c) We also update the residuals: $r_i \leftarrow r_i - \lambda \hat{f}^b(x_i)$.

Finally, the output of the boosted model can be expressed as follows:

$$\hat{f}_{boosted}(x) = \sum_{b=1}^B \lambda \hat{f}^b(x). \quad (2.8)$$

As for the choice of tuning parameters for boosting model, according to James et al. (2013), λ the learning rate are typically 0.01 or 0.001, and $d = 1$ with resulting 2 terminal nodes. However, the selection of the number of tree B is bit complicated since boosting can overfit if B is too large, although the overfitting occurs very slowly. Thus, we used cross-validation technique and selected 54 number of trees for B , which happens to be quite close to the number of tree where the test set error is at its minimum, as illustrated in **Figure2.2**.

As James et al. stated, combining a large number of trees often effectively improves the prediction accuracy, however at the expense of some loss of interpretation. Nevertheless, one can use ensemble tree models, including boosting, to detect the interactions between variables which add predictive value to credit scorecard. Perhaps, such testing of variable importance can be used as a variable selection or reduction method for the large set of predictor variables (refer to **Figure2.3** for illustration).

2.2.3 Neural Network Model

In recent years, the progress on Artificial Intelligence (AI) has been made in attempt replace domain of thinking and decision making with computers. *Neural Networks*(NNs) are the core of AI for the use of predictive model as part of a decision process.

According to Anderson, Neural Networks are describes as:

networks of computing elements that can respond to inputs, and learn to adapt to the environment. NNs do not follow formulaic procedures as other statistical technique would, rather they train through the repetition of examples.

The end result of Neural Networks are something like a decision tree, however the details is much finer and the decision rules are much complex and difficult to interpret. According to Neural (2002) the best suited Neural Network development technique for credit scoring is the Multilayer Perceptron (MLP) as it can deal with both non-linearity and interactions easily. The NNs have the advantages of being able process huge amount of data, discover patterns and relations in the data, and adaptable to changing circumstances. On the other hand, they can overfit to the

training sample and overly complex in a way that the relations detected by the models are very difficult to interpret. Nevertheless, one make a use of such model as a benchmark comparison to other credit scoring models.

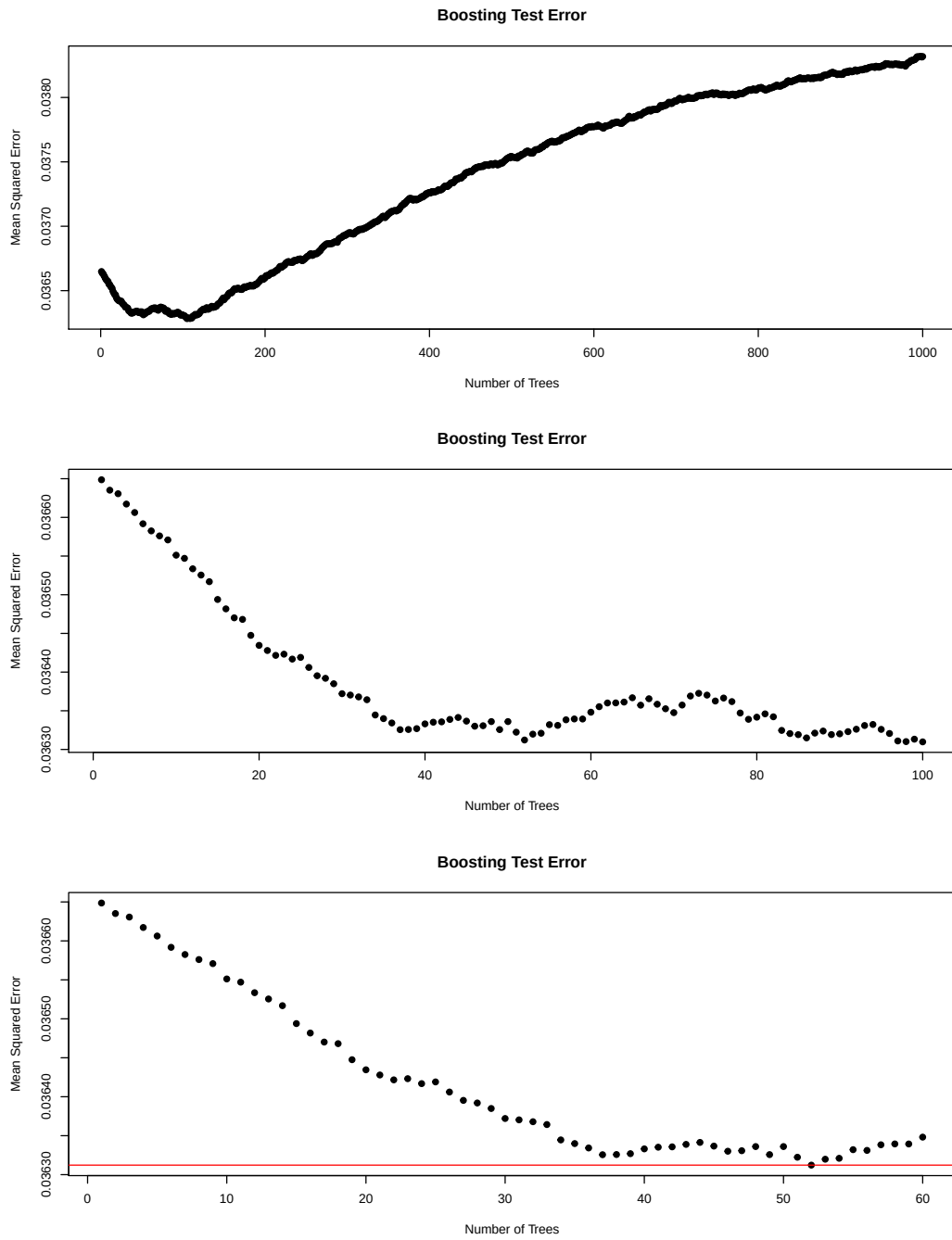


Figure 2.2: Boosting Test Error. The plot shows the MSE relative to the number of trees selected for boosting. The red line represents that minimum MSE at 52 trees.

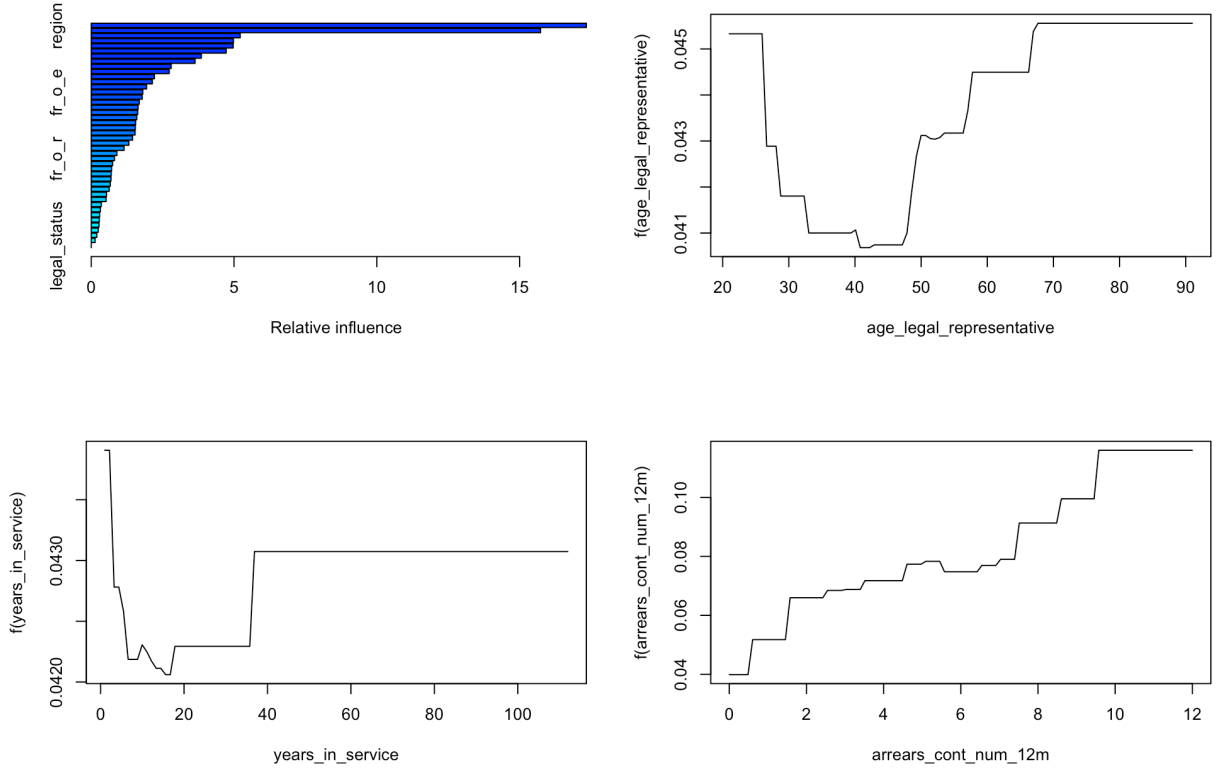


Figure 2.3: Variable Importance and Risk Characteristics based on Boosted model.

The top left plot shows the relative influence of predictor variables on the target variable. The remaining plots are randomly selected illustrations of predictors values relative to their corresponding default risk. Notice that default risk declines for the customers around the age between 20 and 50, and the risk picks up around 60 again. The risk of the newly established business relates to high default risk and risk probability stabilizes for the old business, whereas the risk increases with additional records of arrears in payments.

2.3 Performance Measures

Usually, several statistical tools are used to analyze the result of the scorecard development. In this research we mainly focus on the optimization of the Gini coefficient and Area under the curve AUC. Also we use Receiver Operating Characteristic (ROC) curve as a visualization tool that illustrates the performance of a binary classifier system.

If a default event is correctly classified and predicted as default, we call it true positive (tp); and if a non-default event is wrongly predicted as a default, it's counted as a false positive (fp). Accordingly, the following parameters are computed:

$$rp\ rate = TPR = \frac{Defaults\ correctly\ classified(tp)}{total\ defaults(p)} \quad (2.9)$$

$$fp\ rate = FPR = \frac{non\ defaults\ incorrectly\ classified(fp)}{total\ non\ defaults(n)} \quad (2.10)$$

Plotting the fraction of correctly classified defaults (tp rate) versus the incorrectly classified non-default (fp rate) give rise to the ROC curve. (for illustration refer to **Figure 4.5**)

Often simply referred as AUC, or AUROC, is equal to the probability that a classifier would rank a randomly chosen positive instance higher than a randomly chosen negative one, given that one assumes 'positive' ranks higher than 'negative'.

$$AUC = \int_{-\infty}^{\infty} TPR(T)FPR'(T)dT = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I(T' > T)f_1(T')f_0(T)dT'dT = P(X_1 > X_0)^3 \quad (2.11)$$

where X_1 is the score for a positive instance and X_0 is the score for negative instances.

Once we obtain AUC, gini coefficient can be expressed as follows:

$$Gini\ coefficient = 2 * AUC - 1 \quad (2.12)$$

Table 2.2: Gini Coefficients. Model Performances on Training and Testing Samples: As exhibited, decision tree model based on simple RPA does not perform well, whereas boosted and NN models clearly show an example of overfitting.

Model	Train	Test
Decision Tree/RPA	0.251	0.249
Boosted	0.657	0.226
Neural Network	0.403	0.351

Chapter Summary

- In this chapter, we briefly discuss the main issues concerning scorecard development process.
- We attempt to focus more on the sections on missing value and recursive partitioning algorithm as these topics will be revised in the later chapters.
- We provide some empirical results and figures for illustrative and explanatory purpose. The description on data and sampling can be found in Chapter 4.
- The rest of the work elaborate on logistic regression models based on differing binning algorithms.

³In the left-hand side equation the integral boundaries are reversed as T has lower value on the horizontal axis.

Chapter 3

Automatic Binning Process

3.1 Binning Process

Binning of predictor variables, also referred as coarse classing, is a part of the *variable transformation and selection* process in the model development process, particularly for linear and logistic regression models. Both categorical and numeric characteristics can be binned in a way that continuous variable to be transformed into few discrete intervals and numbers of differing levels in a factor variable can be bucketed into fewer classes. The reason for fewer classes are preferable is that greater the number of bins, greater the potential for complexity and over fitting. At the same time, having too few coarse classes can lead to loss of information value. Thus, the desired property of binning is to find the minimum number of bins with maximum possible information value (refer to the appendix on information value).

Aside from the trade-off between the number of bins and information value, the following "logical" list should be also taken into consideration (Anderson, 2007):

- Logical groups: each groups should be of similar risk or have some logical connection
- Logical breakpoints: breaks define the classes for continuous characteristics and should also make sense
- Logical relationships between groups: risk difference between the classes should make intuitive sense.

In some situations, a monotonic relationship is required between predictors and risk. Such relationship can be enforced through the binning process as well.

3.1.1 Manual binning vs Automated binning

In earlier section, we used manual binning for the logistic regression, where the class bands are predetermined based on the expert knowledge. Manual binning offers detailed and tailored solution to classification within variables in line with business sense. For example, a variable can be binned according to the target group study by the firm, and such binning will make more business sense than some random classification based on equal size or equal weight binning.

Especially for categorical variables, employing local expert knowledge has clear advantage over automated binning. Let's consider a variable for "region" or "province" for instance: local managers have more knowledge on characteristics within the variable, including socio-economic situations and particular legislation in the region etc, and hence result in better/precise binning of the variable by groups of regions with similar characteristics.

However, manual binning is a tedious and time consuming process, especially when the number of predictor variables is large. Perhaps, one may deal with large dimensionality of the data by applying variable reduction techniques mentioned in the earlier sections, however, such process will need to be reviewed and validated later on. Automated binning on the other hand, offers time saving development and validation process, as well as reduced possibility of excluding important variables by a mistake that could derive from various variable reduction/selection techniques.

Therefore, in order to reduce time and effort that LeasePlan analysts spend to develop and validate scorecard models, particularly the tedious binning process, we aimed to assess and improve automated binning algorithms and various binning packages. The *Section 3* of this chapter is on equal width and equal weight based binning. These are the simplest automatic binning techniques and yield unstable results; however, they are good starting point for the further techniques. Then, *Section 4* is on recursive partitioning algorithms for discretization through decision tree models. Finally, in *Section 5* and *6*, we assess the Monotonic Adjacent Pooling algorithm in combination with Cluster analysis. Also we consider the performance automated binning tool in SAS enterprise miner and "Smbinning" R package in **Chapter 4**.

3.2 Simple Binning Algorithms: Equal size and Equal width binning

Equal size or equal width binning methods are the simplest unsupervised discretization techniques which divide a continuous feature into bins without taking into account any other information, such as a relation between the feature and default frequencies. However, one has to predefine the number of bins.

Equal width binning:

The predictor variable domain is divided into equal width, k number of sub-intervals.

$$w = \frac{(Var_{max} - Var_{min})}{k} \quad (3.1)$$

Equal sized binning:

$$w = \frac{length(Var)}{k} = \frac{n}{k} \quad (3.2)$$

Assume n is the length of the variable (number of observation).

In a case where n is not multiple of k , some of the bins contain either $[n/k]$ or $[n/k] + 1$ points. Thus, instead of requiring bins to be equal sized, we can adjust them into similar sized bins. If we assume that $m_1, m_2, \dots, m_p$ are the number of measurements taken at the predictor variable, then for a given number of K bins, we obtain the set of bins $I = (I_1, I_2, \dots, I_k)$ that minimizes

$$\sum_{k=1}^K \left| \sum_{j \in I_k} (m_j - \frac{n}{K}) \right| \quad (3.3)$$

where $n = \sum_{j=1}^P m_j$. (Sonehag et al., 2012)

According to Dinero (1996), these binning techniques are more suitable for analyzing distribution of a variable due to their simplified nature, i.e classic histogram, yet not preferable for binning of credit-scoring models as they don't conciliate the requirements for proper binning mentioned in Section 3.1, such as the logical list and monotonic relation to risk.

3.3 Recursive Partitioning Algorithms

In Chapter 2, we covered a topic on fitting various trees to classify and select predictor variables. We learned that tree based models show modest performance due to their unstable nature. However, we may also use decision tree models, including Classification Regression (CART) models, to categorize a variable into bins mapped to a binary target variable for its ulterior usage in scoring models, such as finding optimal split for a given numeric variable (Jopia, 2016).

So the basic idea behind such method is to grow a classification tree inside the variable itself, obtain terminal nodes based on the predefined splitting criterion, and then group the characteristics together with those who fall into the same end node.

Some statistical packages are available for employing CART models for the purpose of coarse classification, such as SAS enterprise "interactive grouping node" tool and "smbinning" package in R, which we'll discuss the comparisons and performances later in Chapter 4.

3.4 Monotonic Adjacent Pooling Algorithm

Pooling algorithms are routine processes that are designed for automated coarse classing to ensure the minimal information loss. It starts with the fine classing, which requires that the maximum number of classes or groups to be created to represent every characteristics in the variable. Thereupon, the algorithm collapses these fine classes into fewer number of coarse classes.

In general, pooling algorithms can be divided into three main types (Anderson, 2007):

- *Non-adjacent pooling*

Non-adjacent algorithms are for categorical variables as it require no assumptions about the ordering of the characteristics. The process collapses pair of classes with least difference in their information values, and would then be repeated for the remaining attribute groups till the optimal groups are obtained.

- *Adjacent pooling*

The algorithm can be applied to numeric variables only, i.e ordinal, discrete or continuous. It collapses the pair of neighbor classes where the least information is lost and continues the iteration till the optimal groups are formed.

- *Monotone adjacent pooling*

This algorithm builds on adjacent pooling, while ensuring monotonicity in the resulting coarse classes.

Our further research is mainly focused on Monotonic Adjacent Pooling Algorithm (MAPA). The reason is that we assume monotonic relationship between our predictor variables and the default risk. For example, let's consider a variable "income"; ideally, the bad rate would go down as the characteristic value increases. So, we start at the lowest characteristic value and keep adding values until the cumulative bad rate hits its maximum. This is the first course classification point. Then, we start calculating the cumulative bad rate from this point until it again hits

a maximum, which brings us to the second break point. Such process is repeated until all the break points are obtained (Thomas et al., 2002). The algorithm can be demonstrated by two sets of equation, "Cumulative Bade Rate" and "MAPA":

$$1. \text{ Cumulative Bad Rate} = C_{k,v} = \frac{\sum_{i=1+V_{k-1}}^v B_i}{\sum_{i=1+V_{k-1}}^v (B_i + G_i)}$$

where V is a vector which contains the series of break points. Since i and k are indices representing each class and each break point, the cumulative bad rate is calculated for all classes above the last break point (denoted as v).

Then, the class with the highest cumulative bad rate is identifies and assigned to the vector as follows:

$$2. \text{ MAPA} = V_k = \max\{v | C_{k,v} = \max\{C_{k,v}\}\} \text{ for all } v > V_{k-1}$$

This iterative process is repeated until the maximum cumulative bad rate is the one associated with the highest possible characteristic value. (Anderson, 2007)

3.5 Cluster Binning Algorithm for Categorical Variables

In previous section, we covered Monotonic Adjacent Pooling Algorithm for coarse classification of numeric variables. When it comes to binning of categorical variables, one may consider Non-adjacent pooling algorithm. ***Alternatively, we propose a method for binning of categorical variables, based on clustering of fine classes.***

Before we go into the details of factor variable transformation based on clustering of homogeneous groups, we provide the following basic concepts and methods on *cluster analysis* :

First of all, cluster analysis is an unsupervised learning method. Supervised learning methods, including regression problems and predictive models centers on the predictive variable. Such models have prior specified labels of the classes. Under unsupervised learning, however, the class labels are unknown and the classification is based on the observed and tested similarities among the observation.

Principal component analysis and cluster analysis are one of the most widely applied methods of the unsupervised machine learning.

In contrast to Principal component analysis which finds the low dimensional representation of the data, cluster analysis looks for homogeneous subgroups of the observations. In other words, instead of looking for the variance, it looks for similarity among observations. Thus, the task of

segmenting the similar kinds of population is a key application of clustering. *Such unlabeled segmentation is what we seek for automated binning of categorical variables, given the limited access to local expert knowledge.* In the case of LPIT data, for example, the attributes of factor variables are named in Italian and consist of more than 20 levels, thus manual binning would be challenging and time-consuming for non-local scorecard developer. Moreover, non-adjacent pooling algorithm is also not suitable for the data, since it requires extra consideration to be taken to ensure that the resulting bins are logical. The two most popular methods of clustering are:

K-Means clustering We seek to partition the observations into a pre-specified number of clusters. Assume that C is the sequence of subgroups that partitions the observations. The sequence has the following property:

$$C_1 \cup C_2 \cup C_3 \cup \dots C_k = \{1, \dots, n\} \quad (3.4)$$

where each observations belongs to at least one of the K clusters. But note that the clusters are non-overlapping, $C_k \cap C_{k'} = \emptyset$, meaning no observation belongs to more than one cluster.

The optimum clustering can be defined in a way that the within-cluster-variation is as small as possible, meaning the observation points in the same cluster groups are closer together. Therefore, over the scope of the series C_k which contains the subgroups of partitions, we want to minimize the total within-cluster-variation, summed over all K clusters. This statement can be formulated as follows:

$$\text{minimize}_{C_1, \dots, C_k} \left\{ \sum_{k=1}^k WCV(C_k) \right\} \quad (3.5)$$

where $WCV(C_k)$ is Euclidean distance:

$$WCV(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^p (x_{ij} - x_{i'j})^2 \quad (3.6)$$

, note that $|C_k|$ represents the number of observations in the k th cluster.

The general algorithm can be expressed as follows (James et al., 2013):

1. Randomly assign the observations into K clusters. These serve as initial cluster assignments for the observations
2. For each of the K clusters, compute the cluster centroids. Centroid is the vector of the p feature means for the observations in the k th cluster.
3. Reassign each observation to the cluster group whose centroid is closest, based on Euclidean distance.

The process 2 and 3 will continue iterate until no further improvement can be made for the cluster assignment.

Hierarchical Clustering This method has advantage over K-means clustering, as it does not require the we commit to a particular choice of K bins.

Agglomerate clustering: Starts from the bottom individual observations as leaves and combines them successively in a hierarchy.

The general algorithm can be expressed as follows (James et al., 2013):

1. Start with each observation, identify the closest two points and merge them.
2. Repeat the process till all points are in a single cluster.

In other words, compared to K-mean cluster algorithm which randomly assigns observations into K clusters and reassign till the optimal end, hierarchical cluster algorithm starts with the individual observations and group them till they all belong to a single cluster or till the threshold. For better illustration, we provide the following graph:

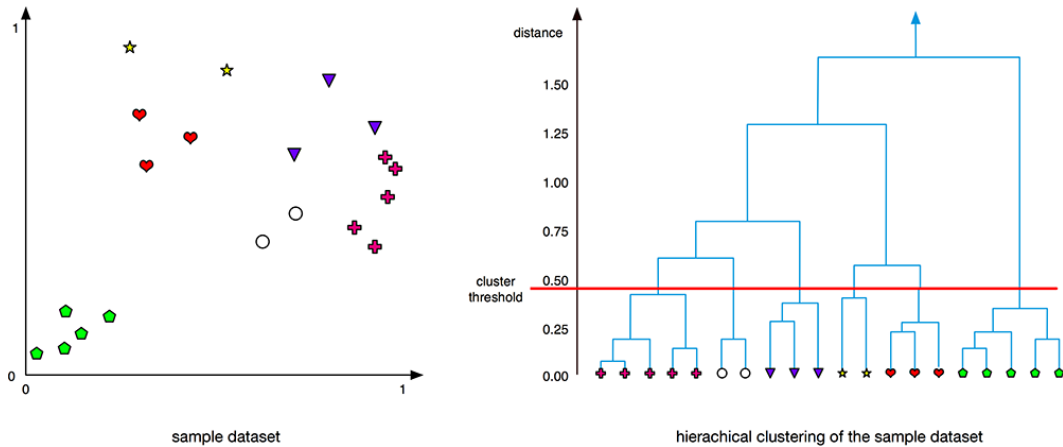


Figure 3.1: Hierarchical Clustering¹.

Hereupon, we end the session on background information and basic concepts of cluster analysis. In the following section, we'll go into the main topic of cluster binning algorithm and introduce our proposal on exploiting the unsupervised learning technique for factor variable classification.

¹Source: <https://www.biostars.org/p/201214/>

3.5.1 WoE based Cluster Binning Algorithm

As one may have noticed, cluster analysis centers around reducing the Euclidean-distance of within cluster variation of observations, while maximizing the segregation among the different cluster groups/bins. However, the Euclidean distance is the "ordinary" distance between two points. To solve the problem, we propose to cluster the factor observations based on their Weight of Evidence (WoE).

WoE is widely used in credit scoring to separate good accounts and bad accounts. It compares the proportion of good accounts to bad account at each attribute level, and measure the strength of the attributes of an independent variable in separating good and bad accounts. As we cluster the factor attributes based on their corresponding WoE, the resulting bins would be analogous to default risk, regardless the fact that unsupervised learning methods don't take any response variable into consideration.

This method can be summarized as follows:

1. Count the number of default and non-default events for each attribute levels in the factor variable.
2. Calculate WoE of each levels of attributes (WoE for the fine classes).
3. Cluster the attributes based on their calculated WoE.
4. Form the final bins (coarse classes) according to the cluster groups.

Although the method of clustering based on WoE has not been proposed yet (or scarce documentation can be found if any), the practice of combining categories with similar WoE is common for scoring models. It is because the categories with similar WoE usually have similar proportion of events and non-events, and thus we may assume the the behavior of the categories is same.

The performance of this proposal method, together with the other automated-binning methods, will be discussed in the following chapter.

Chapter 4

Empirical Analysis and Results

4.1 Data Description and Sampling

In this research, we use data provided by LeasePlan entity in Italy (LPIT). The client portfolio mostly consists of retail businesses and Small Medium Enterprises (SMEs), and the observation period starts from 2009 till 2013.

The data set was extracted from several sources and internal data warehouses, which contain the following information:

- Client information (Client ID, age, years in service,...)
- Product information (number of the leased cars, value of the car/contract,...)
- Financial information (financial ratios, income, revenue,...)
- Internal behavioural information (number of arrears and delinquency records,...)
- External information (credit bureau score,...)
- Macroeconomic indicators (GDP and unemployment rate by geographical areas,...)

As for the default definition, LPIT adopts policy of minimum 90-days past due. The status of client is monitored over 12 months after the date of delivery of the first vehicle related to the contract.

In total, our data set contains 5860 observations, 47 predictor variables and a binary flag variable for default instances. Out of all the observation, there are only 213 entries of default events recorded. We partition our data set into simple random samples of 70% observations of training set and 30% observation of validation set. The training and validation samples each contain 146 and 67 defaults, respectively.

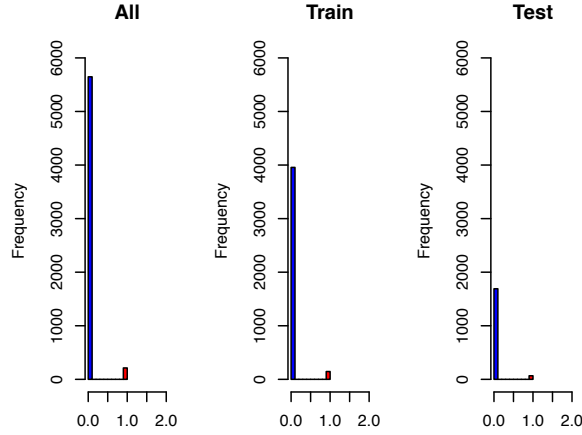


Figure 4.1: Frequency Histogram of flag variable

4.2 Initial Variable Selection

Univariate Analysis Initial stage of model development process involves in analyzing distributions of predictor variables and checking for outliers and missing values. Most of the variables had moderate level of missing values (refer to **Figure 4.2**). We excluded 3 variables from the dataset, which had more than 50% of their observations missing. However, we attempt to be careful with decisions to drop variables based on missing values. That is because for some variables missing values may imply a cause, such as missing values in variable "arrear in 9 months" means that some clients don't have any arrears or delinquency record.

Distribution Analysis: Perhaps the simplest way to determine characteristics of a variable, such as whether it's continuous, discrete, or categorical, is through the examination of its distribution. Visual approaches like histogram, box-plots, cumulative distributions and QQ-plots are commonly used. For example, one can observe from **Figure 4.3** that variables like "credit bureau score" and "arrear in 6m" appear to be discrete, whereas "age" and "delinquency rate in 6m" appear to be relatively continuous. This kind of visual approach might also offer possible binning suggestions, since discrete variables are most likely to be split by their differing levels (i.e credit bureau score).

Moreover, one can detect outliers in the variable by examining its distribution, such as calculating its mean, median and quartiles (at 99% etc.). For instance, "Financial Ratio1" has several outliers at values of 2000 and 4000, whereas most of its observations fall between $[-1;1]$.

In most cases, outliers can be solved through proper coarse classification, where the extreme values are grouped into the first or the last bin classes. However, one may need to correct for outliers when using binning approach that are sensitive to outliers, such as Equal-width binning which divides the variable domain ($Var_{max} - Var_{min}$) into sub-interval.

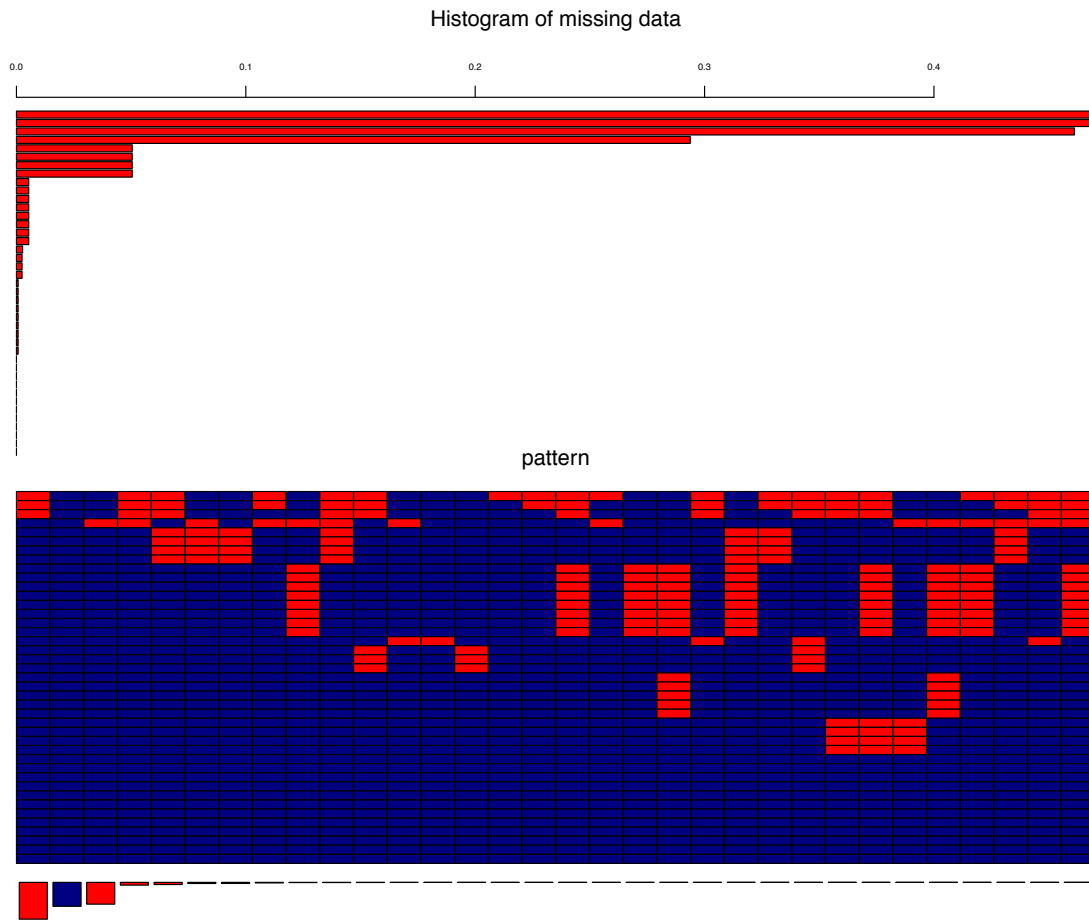


Figure 4.2: Missing values in the full data set. For confidentiality purpose, the name labels are invisible.

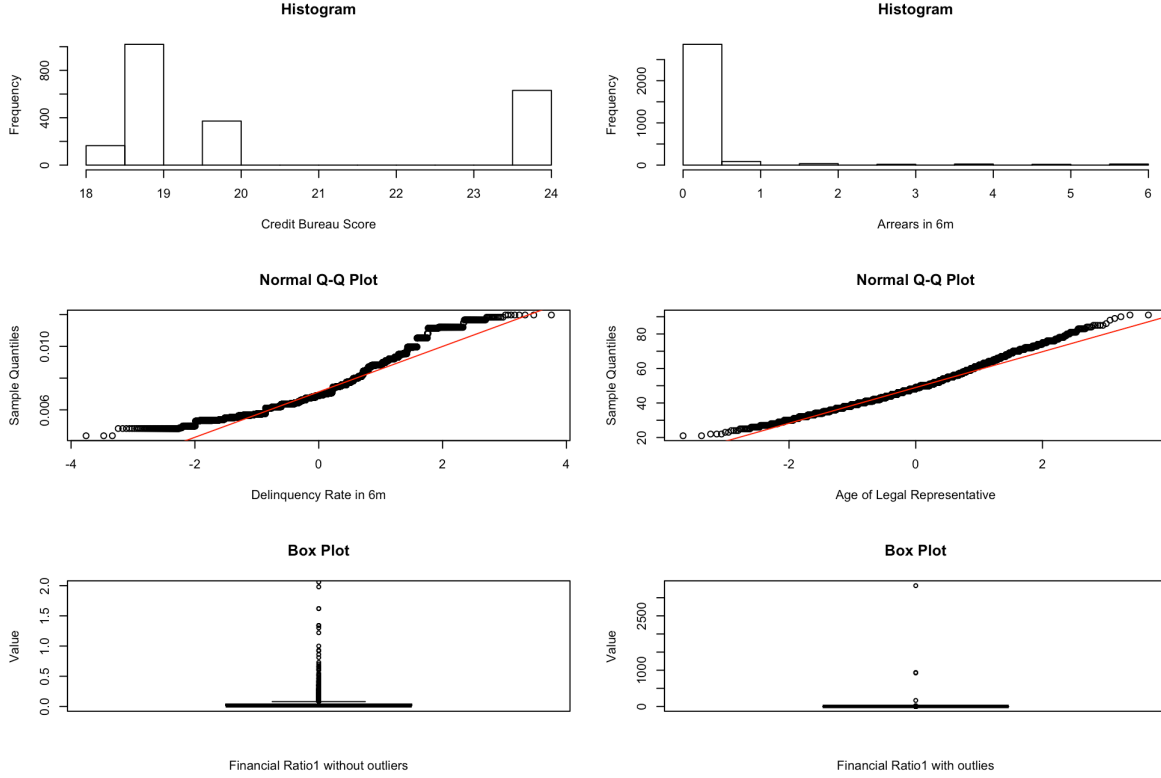


Figure 4.3: Visual Illustrations for Randomly Selected Variables

4.2.1 Automated Binning Models

In Chapter 3, we introduced concepts and methodologies for automated binning algorithms. Here, we compare and examine the logistic regression models where the variables are binned based on those differing automated binning algorithms. We also examine the performance of available statistical packages for automated binning processes. The models are illustrated as follows:

- Model1: **MAPA+Clust**
- Model2: **RPART**/Recursive Partitioning
- Model3: **CTree** /Recursive Partitioning
- Model4: **SAS** /Monotonic Event Rate

As a benchmark, we compare the models with manually binned model logistic regression model (developed by LPIT).

MAPA+Clust The model combines Monotonic Adjacent Pooling algorithm for numeric variables and Cluster binning algorithm for categorical variables. We used R package "Ckmeans.1d.dp"¹ for clustering of the individual variables, as the package is designed for 1-dimensional data.

RPART The model exploits recursive partitioning for CART models to obtain optimum split points and groups the observations based on the terminal nodes of the individual variables. This automated binning process can be executed using R package "Smbinning", which utilized another package "Rpart" for recursive partitioning.

CTree In this model, we improved the "Smbinning" package by replacing "Rpart" package by "Party" package of Conditional inference trees (CTree). According to Hothorn et al. (2015), CTree offers relatively unbiased recursive partitioning based on permutation tests, compared to RPART. We also altered "Smbinning" package to allow the user to set initial definitions, such as minimum bucket size, minimum splits, inference level etc.

SAS The model bins are arranged according to "Interactive Binning Node" tool in SAS Enterprise Miner. SAS is a commercial statistical software and "Interactive Binning Node" tool has three options for automated binning: "Tree based method", "Ordinal Grouping method", and "Monotonic Event Rate". The first two methods provided similar results to "Smbinning" package, thus we used "Monotonic Event Rate" method as the main comparison to our "MAPA+Clust" model.

Model Comparison: Binning of Categorical Variables

As our main proposal in this research is cluster binning method, we paid detailed attention to coarse classification of the categorical variables. Here below, one can observe the differences in number of bins of categorical variables determined by the 4 models:

Var Name	# Levels	MAPA+Clust	CTree	Rpart	SAS
Area	6	3	2	1	4
Region	21	5	2	1	5
Branch Code	25	5	2	1	5
Legal Status	5	2	1	1	2

Table 4.1: Number of coarse classes of categorical variables. Note that the second column represents the initial level of attributes in each variables.

¹refer to appendix for the package details

It appears that the worst performing model is "RPART" as it doesn't find any optimum splitting points among observations/attributes. "CTree" model also provides modest results, especially for the factor variables with large number of levels, i.e. "Region" and "Branch code". However, the reason that recursive partitioning models perform poorly can be due to the low quality data set, where the number of default events recorded is small compared to the number of total observations.

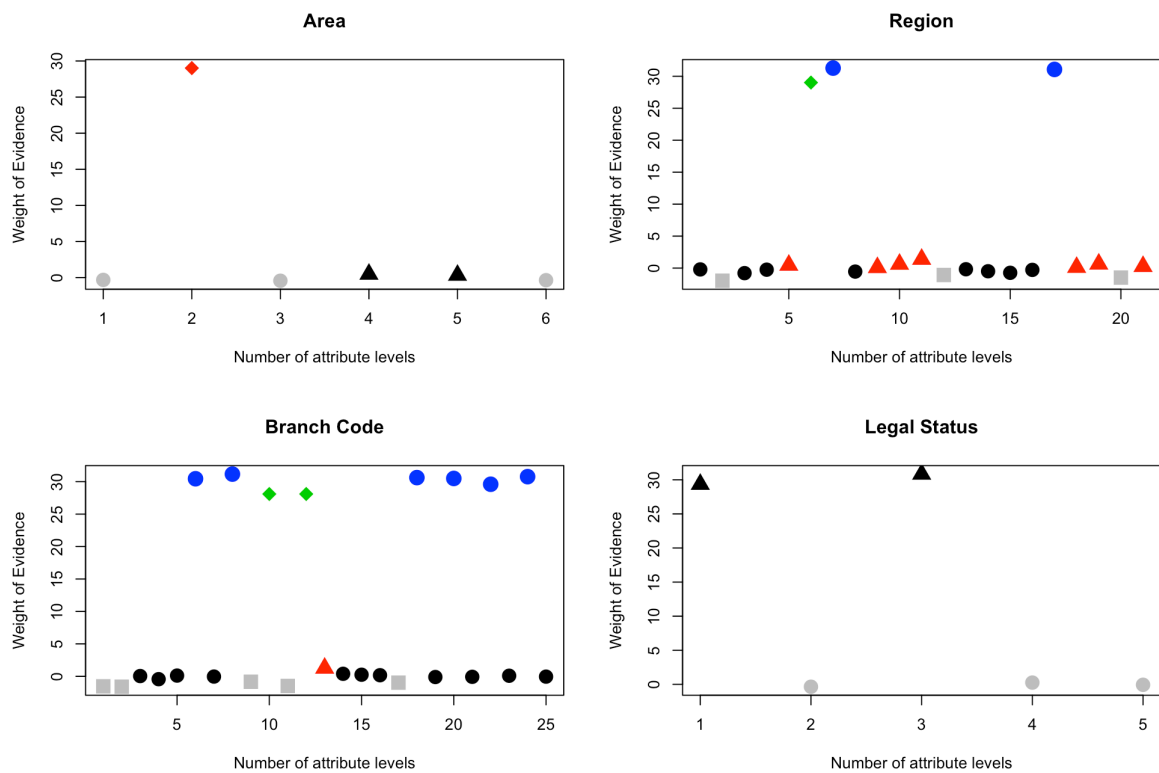


Figure 4.4: Clustering of Categorical Variables. Attributes are grouped into clusters with those who are of similar level based on their calculated WoE. The symbols shows which attributes are in the same cluster bin.

Model Comparison: Variable selection based on IV statistics

Information Value (IV) statistic is a widely used method for selecting predictor variables for binary logistic regression (Lund and Brotherton, 2013). In most of literatures on credit scoring, they use IV value of 0.3 as cutoff point, based on IV Rule of Thumb for evaluating the strength of predictor variable. (see Finlay, 2010; Siddiqi, 2006)

IV Rules of Thumb:

- Less than 0.02 : Not predictive
- 0.02 to 0.1 : Weak
- 0.1 to 0.3 : Medium
- Greater than 0.3: Strong

However, the optimum IV cutoff point for variable selection shall be determined based on the data set and the number of available predictor variables. Moreover, IV value of the individual variable depends on the way the it's binned, such that greater the number of bin classes, greater the IV-value.

Based on the average predictive performance of 44 individual variables, we set the variable selection cutoff point at **IV of 0.1**, as illustrated in table below.

Table 4.2: Summary of IV statistics of predictor variables.

	MAPA+Clust	CTree	Rpart	SAS	Manual
Mean	0.159	0.106	0.051	0.274	0.128
St.dev	0.177	0.103	0.064	0.324	0.103
Max	0.950	0.321	0.175	1.124	0.321
Min	0.00	0.00	0.00	0.046	0.00

Out of 44 predictor variables in the data set, excluding the 3 variables dropped due to high percentage of missing values, we select 25, 18, 9, 24 number of variables for "MAPA+Clust", "CTree", "Rpart", and "SAS" models, respectively.

4.3 Final Variable Selection

In previous section, we focused on preliminary variable selection based on distributional aspects and individual predictive power, i.e $IV \geq 0.1$. Once the long list of variables are selected, one should further assess the multicollinearity among the variables for the final model.

Depending on the abundance of pre-selected variables, one may consider reducing the number of numeric variables based on correlation or cluster analysis. For example, a common practice is to drop highly correlated variables in their order by IV, perhaps when the correlation coefficient is more than 0.5. Similarly, one can assign the variables into either disjoint or hierarchical clusters on a similarity measure. Variables in the same cluster group are often highly correlated with each other but are less correlation with variables that belong to other cluster groups. Hence, one can select one or two variables from each clusters with minimum R-square² ratio within its own cluster (Aggrawal and Kosain, 2011).

As we take into consideration that the number of variables selected by our models are small, we decided to employ stepwise logistic regression without taking any prior variable elimination measurements. Stepwise regression is carried out by an iterative process, either backward elimination or forward selection of variables, or bidirectional elimination. We decided to employ **forward selection** iteration, which starts with no variables in the model, adding a single variable at each steps using a chosen model comparison criterion. Predictor variables are added in their order of relative improvements to the model and the process is repeated until none of the remaining variable improves the model.

The process can either take form of a sequence of F-test, t-test, Adjusted R square, Akaike information criterion (AIC), or Bayesian information criterion (BIC) and so on. We utilize Akaike information criterion and Bayesian information criterion for the final variable selection of our models based on stepwise forward logistic regression.

4.4 *Performance Evaluation: Testing for Discriminatory Power*

Once logistic regression models are built for the proposed automatic binning algorithms trained on the development set, we assess the model performances on the testing sample for validation purpose. Widely used statistics like Kolmogorov Smirnow chart and Gini coefficients are typically computed for the purpose of evaluating the improved odds for differentiating the good credit applicants from bad credit applicants (Thomas et al., 2002).

² $RSquareRatio = \frac{1-RSquareOwnCluster}{1-RSquareNextClosest}$

In the table below, we provide Gini coefficients of the model performances on training and test/validation samples:

Table 4.3: Gini Coefficients Note that the first column represents the automated binning models, the second column represents the stepwise regression models evaluated on different criterion. Both "gini_Foward" and "gini_BIC" are forward selection models based on AIC and BIC respectively. "gini_Full" is simple logistic regression of all the pre-selected variables.

		Train	Test
MAPA+Clust	gini_BIC	0.505	0.349
	gini_Foward	0.506	0.348
	gini_Full	0.535	0.364
CTree	gini_BIC	0.420	0.283
	gini_Foward	0.435	0.295
	gini_Full	0.455	0.294
Manual	gini_BIC	0.470	0.300
	gini_Foward	0.499	0.336
	gini_Full	0.502	0.341
SAS	gini_Foward	0.833	0.091

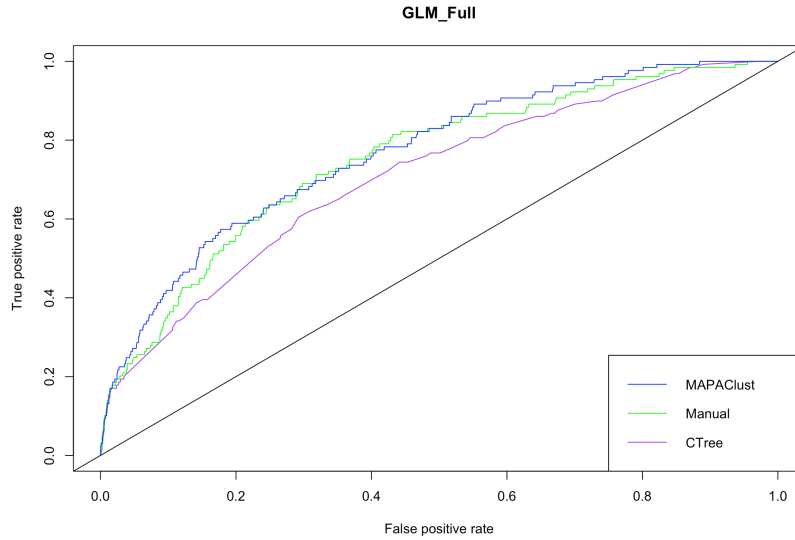


Figure 4.5: ROC Curves

As one may observe, the predictive power of the models declines when applied to the test/validation sample. Especially, SAS/monotonic event rate based model has clearly over-fitted to the training sample, and as the consequence it performs poorly on the test sample.

4.5 Model Selection and Assessment

Scorecard development is an iterative process and the resultant scorecard must satisfy a number of performance standards, including:

- Stability - The scorecard attributes should be estimated from a sufficiently sized dataset that covers appropriate historical period.
- Discriminative - The scorecard is expected to distinguish between the goods and bads.
- Interpretable - The output of the scorecard should be understandable and explainable to non-experts.
- Not overly complex - The scorecard should not-rely on any single feature. Also, it should not consist of too many features.

On the other hand, there is no quantitative definitive framework available to ensure that the criterion is satisfied. The decision making process varies for each organization, depending on a range of factors such as the available data and resources, as well as the corporate and regional cultures. In Chapter 2, we briefly assessed suitability of popular state-of-art models for scorecard modelling. For the sake of interpretability and simplicity, we decided to elaborate on logistic regression approach by exploring several binning algorithms.

Among the choice of four models built on different automated binning algorithms, we found that models based on recursive partitioning algorithms are not suitable for our data, as they don't find optimal splitting points for coarse classification of predictor variables; although "CTree" model, which is based on conditional inference tree, performs slightly better than "RPART" model which is based on simple classification regression trees (CART).

For this reason, we had to narrow down our choices to "SAS" and "MAPA+Clust" models. "SAS" model has the highest gini-coefficient when evaluated on the training sample. However, without additional touch up (manual binning performance) to monotonic event based setting of SAS tool for automated binning, it does not account for ordering and adjacency of ordinal variables. As the result, the model overfits to the training sample and performs poorly on validation sample.

Therefore, we select "MAPA+Clust" model, given that the model effectively bins numeric predictors based on Monotonic Adjacent Pooling algorithm and factor variables based on cluster binning algorithm. Moreover, the model shows slightly higher performance when benchmarked against current LPIT model, where the predictor variables are manually binned based on local expert knowledge.

Chapter 5

Conclusion

The purpose of this work was to find suitable modelling approaches for developing application scorecards. To do so, we review and assess widely used traditional and state-of-art modelling techniques, both theoretically and empirically.

We started our research by emphasizing basic concerns in scorecard development process, such as requirements for the construction of a good quality data set. We spent a great deal of time on investigation of proper approaches to handling missing values, as our data set suffered from significant amount of missing observations with underlying MNAR mechanism.

Some researchers, including Baesens et al. (2003) and Allen et al. (2003), have acclaimed that the area of retail credit risk modelling remained underdeveloped in terms of adapting to newer techniques, i.e cut-edge technologies. Therefore, we briefly assessed the performances of simple decision tree, ensemble of tree, and neural network models for credit scoring, in *Chapter 2*. It's important for an application scorecard that the outcome is interpretable and the decision logic is understandable to clients. Due to their lack of interpretability or poor performance, we consider that such models are not suitable for application credit scoring.

In *Chapter 3*, we focus on binning process and selection of predictor variables. Binning or coarse classification is a part of logistic or linear regression modelling approaches. When manually handled, binning is a tedious and time consuming process; and information loss can often occur from an attempt to reduce the number of variables to be binned, through various variable reduction techniques. Thus, we reviewed several binning algorithms and automatic binning tools.

We developed 4 logistic regression models based on different automated binning algorithms. In *Chapter 4*, we compared the models based on their discriminatory power. As the results suggest, we consider that "MAPA+Clust" model, which is based on Monotonic Adjacent Pooling Algorithm in combination with cluster analysis, is our best model.

We believe that our best model managed to achieve following objectives:

- As we employed logistic regression approach, the model is simple and interpretable.
- The binning algorithm effectively bins both numeric and factor variables.
- Automatizing the binning process reduces time and effort to be spend on development and validation of the model.

Of particular interest to this research are the modelling techniques and the optimization of discriminatory power for credit scoring models. Therefore, the further validation and monitoring stages are not be investigated in this work.

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Appendix A

Equations & Terminologies

A.0.1 Weight of Evidence

Weight of Evidence (WoE): The basic idea behind WoE is to convert the risk associated with a decision onto a linear scale that is easier to assess. The equation can be expressed as:

$$WoE_i = \ln \left(\left(\frac{N_i}{\sum N} \right) / \left(\frac{P_i}{\sum P} \right) \right) \quad (\text{A.1})$$

where N stands for non-occurrence (negative) and P stands for occurrence (positive).

A.0.2 Information Value

Information Value (IV): Related to the sum of the (absolute) values for WoE over all groups. The equation can be expressed as:

$$IV = \sum_{i=1}^n \left[\left(\frac{N_i}{\sum N} - \frac{P_i}{\sum P} * WoE_i \right) \right] \quad (\text{A.2})$$

Appendix B

R Packages

B.0.1 RPART

RPART package:

This package implemented in R uses recursive partitioning algorithm based on a CART (Classification and Regression Trees) model using a two stage procedure.

First, the best variable which partitions the data into two parts is found. Such partitioning process is applied to each sub-groups separately and recursively until the subgroups either reach the predefined minimum size (minimum of 5 observation in each sub-groups, etc) or until no further improvement can be made.

Second, the model built in the first stage is trimmed backwards using cross-validation. In this procedure, a cross validated (default) risk estimate is computed for a nested set of sub trees, such that the final model is the sub tree with the highest estimate of risk.

Compared to "TREE" and other similar packages, it offers superior solution to the missing values in the data set, as it creates additional category of "NA" for both the discrete and continuous predictors.

B.0.2 CTree

PARTY package:

This package utilized recursive partitioning algorithm with implementation of conditional inference trees procedure. RPART package, which employs the CART algorithm, perform an exhaustive search over all possible splits maximizing an information measure (based on IV or Gini) of node impurity selecting the covariate showing the best split. Such approach has two major issues of over fitting and a selection bias towards covariates with many possible splits. Party package overcomes these problems through the implementation of permutation test developed by Strasser and Weber (1999), which is the conditional distribution of statistics measuring the

association between responses and covariates (Hothorn et al, 2006).

B.0.3 Smbinning

Smbinning Package:

This package is developed for automatic binning purpose. It categorizes a numeric variable into bins and calculates metrics such as Weight of Evidence and Information Value (IV). The package also allows customized binning and generates graphical illustrations. Disadvantage of the package is that user can't definite min/max number of buckets or min/max observations in the bucket. User can define the parameter p of percentage of record per bin.

B.0.4 Optimal K-Means Clustering

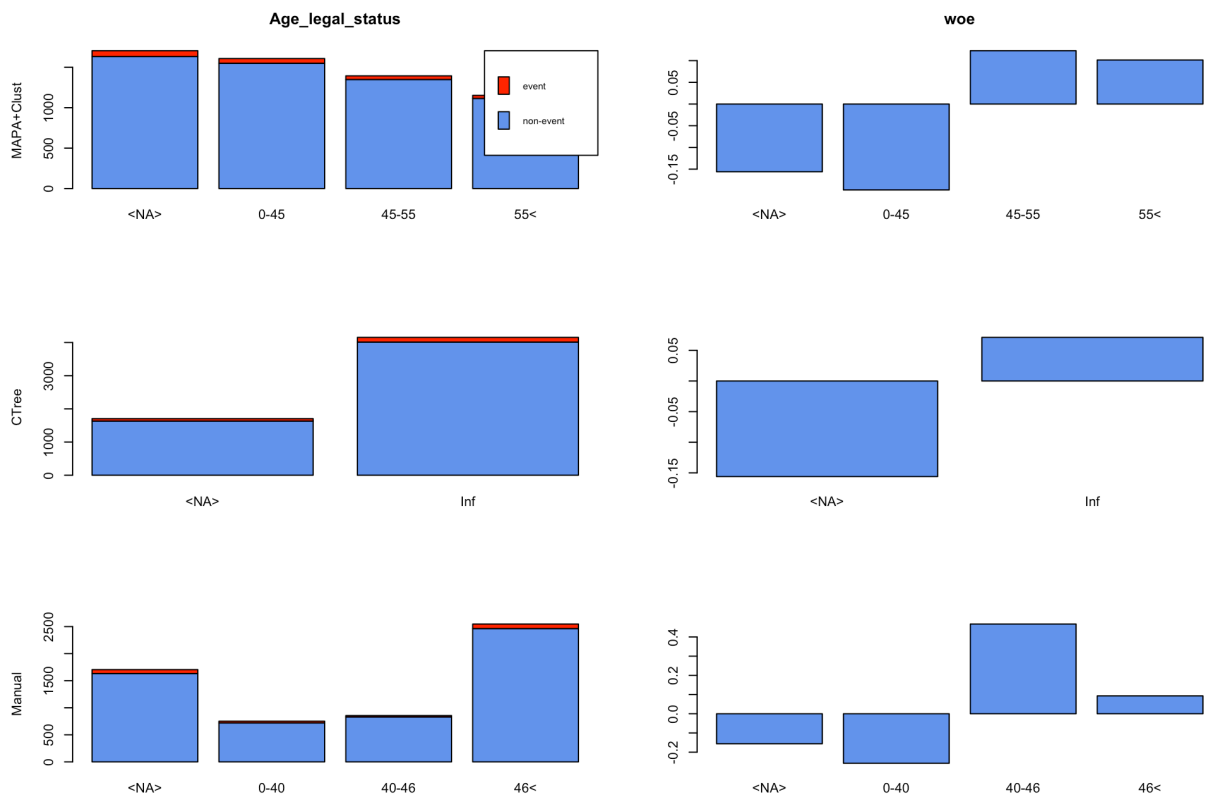
Ckmeans.1d.dp package:

This package utilized dynamic algorithm for optimal K-means clustering. It's particularly suitable for clustering of 1-dimensional data. According to the authors, the package has an advantage over simple hierarchical clustering and heuristic K-means algorithm in terms of efficiency and accuracy in selection of K for univariate clustering. The package returns optimal K parameter based on cross-validation, but user can define the number of K as well.

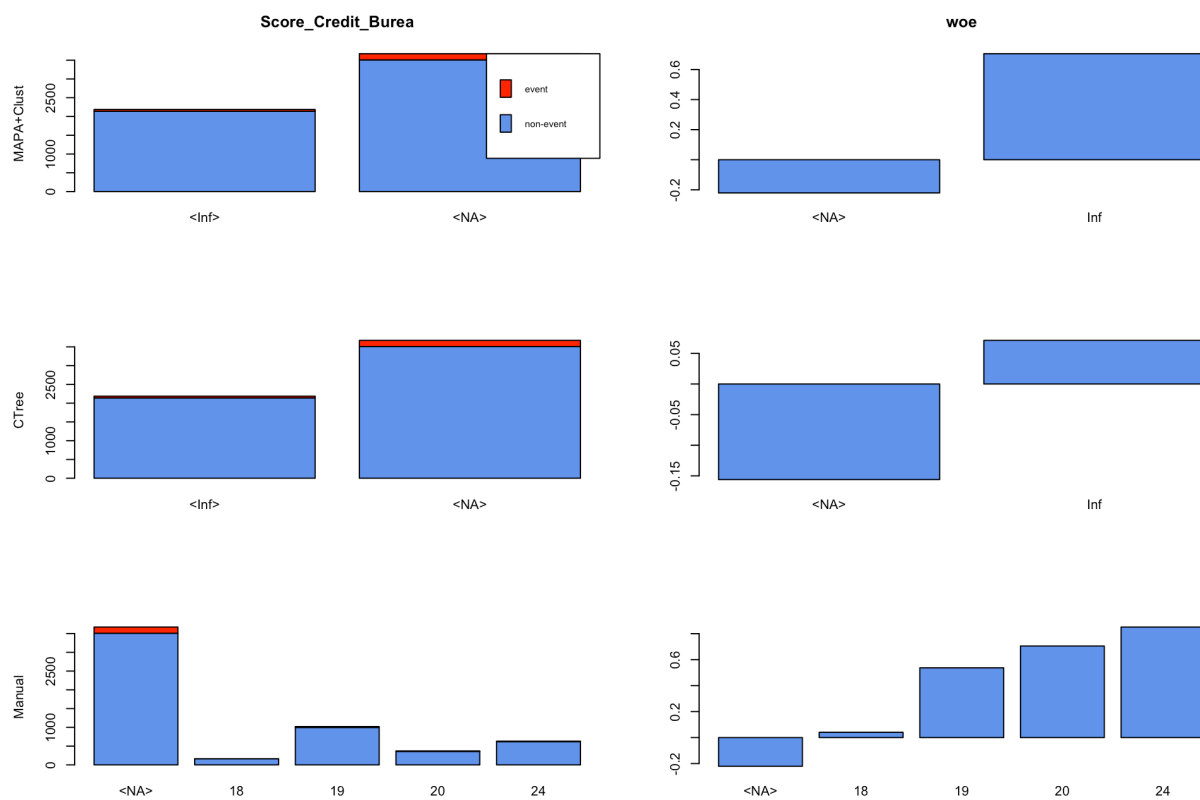
Appendix C

Additional Graphs

C.1 Age_legal_representative



C.2 Score_credit_bureau



C.3 Delinquency_Rate_1Quarter

